

SIMULTANEOUS

EQUATIONS

- 1 Three salad rolls and four pies cost \$20.56 altogether. A salad roll costs \$1.16 more than a pie. How much does each cost?
- 2 The sum of x and y is 12. When y is subtracted from $3x$ the result is 16. Find x and y .
- 3 The sum of two numbers is 38 and their difference is 24. What are the numbers?
- 4 Two books have a total of 500 pages. If the number of pages in one book is 12 pages less than 3 times the number of pages in the other, how many pages are there in each book?
- 5 A farmer uses 140 m of fencing wire to enclose a rectangular paddock. If the length of the paddock is 10 m longer than the width, find the dimensions of the paddock.
- 6 After buying a book, Libby received 18 coins totalling \$1.65 in change. If she was given only 5- and 10-cent coins, how many of each did she receive?
- 7 Five less than twice Michelle's present age is equal to Yesmin's present age. Two years ago, twice Yesmin's age was the same as 10 years more than triple Michelle's age. Find the present ages of the two girls.
- 8 Wes and Tony live in two cities 360 km apart. They leave their houses at the same time by car and travel towards each other at 80 km/h and 100 km/h, respectively.
 - (a) Write an equation for each journey representing the distance from Wes's house (D) at any time (t).
 - (b) How much time will pass before they meet on the highway?
- 9 It takes a barge 3 hours to travel 27 km upstream against the current. On its return journey, with the current, the barge can travel the same distance downstream and an additional 3 km in only 2 hours. Find the speed of the boat and the current.